

Experiment No.	7
Experiment Title	Dimensionality Reduction and Model Evaluation (With and Without PCA)
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1 Objective

To systematically implement, evaluate, and compare the empirical effects of dimensionality reduction using **Principal Component Analysis (PCA)** on the classification performance, stability, and computational complexity of ten distinct machine learning classifiers on the **Wisconsin Diagnostic Breast Cancer (WDBC)** dataset:

- Training and validating models **without PCA** (original 30-dimensional standardized feature space).
- Training and validating models **with PCA** (reduced orthogonal feature space retaining $\geq 95\%$ cumulative variance).
- Executing systematic hyperparameter tuning and 5-fold stratified cross-validation under both configurations.
- Analyzing the effect of dimensionality reduction on variance across validation folds, multi-collinearity mitigation, decision boundary stability, computational speed, and generalization capability.

2 Problem Statement

Automated computer-aided diagnosis (CAD) of breast tumors from digitized Fine Needle Aspirate (FNA) cell nucleus imagery is a critical binary pattern recognition task. The WDBC dataset features 30 continuous numerical attributes describing nuclear morphology. These attributes exhibit extreme pairwise multi-collinearity (e.g., radius, perimeter, and area).

- **Input Space:**

- **No-PCA Setting:** Full continuous feature vector $\mathbf{x} \in \mathbb{R}^{30}$ comprising 10 morphometric properties (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension), each quantified across mean, standard error (SE), and largest (“worst”) measurements.
- **With-PCA Setting:** Orthogonal projected coordinate vector $\mathbf{z} = \mathbf{W}_k^T(\mathbf{x} - \boldsymbol{\mu}) \in \mathbb{R}^{10}$ capturing 95.21% cumulative variance while eliminating inter-feature correlation.

- **Output Space:** Binary diagnostic class label $y \in \{0, 1\}$ denoting Benign (0) or Malignant (1).
- **Optimization Objective:** Maximize classification Accuracy, Precision, Recall (Sensitivity), F1-score, and ROC-AUC under 5-fold cross-validation while investigating the impact of linear subspace projection on ten diverse classifier paradigms.

3 Dataset Description

Attribute	Specification
Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC) Dataset
Dataset Source	UCI Machine Learning Repository [1]
Number of Samples	569 instances
Number of Features	30 continuous numerical features + 1 binary target class
Feature Breakdown	10 Mean metrics, 10 Standard Error (SE) metrics, 10 Worst metrics
Class Distribution	Benign (B): 357 (62.74%), Malignant (M): 212 (37.26%)
Missing Values	0 missing entries across all 30 feature dimensions
Train-Test Split	80% / 20% (455 training samples / 114 test samples, Stratified, <code>random_state = 42</code>)
Feature Scaling	Standardized using Z-score <code>StandardScaler</code> ($\mu = 0, \sigma = 1$)

Table 1: WDBC Dataset Characteristics and Experimental Configuration.

4 Exploratory Data Analysis & PCA Dimensionality Analysis

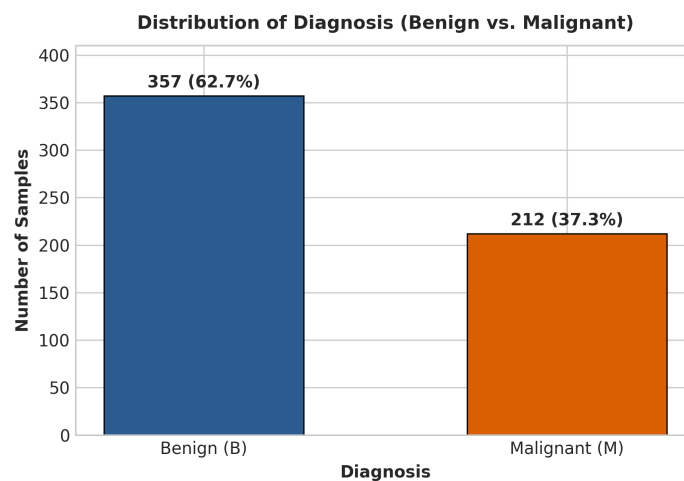


Figure 1: Class Distribution of Benign (B) vs. Malignant (M) samples in the WDBC dataset.

Inference: The target variable exhibits a natural, moderately balanced class distribution with 357 Benign instances (62.74%) and 212 Malignant instances (37.26%). The class ratio is approximately 1.68 : 1, meaning minority class re-sampling is unnecessary. Stratified 5-fold cross-validation is employed to ensure that every fold maintains this exact class ratio, making Accuracy, F1-Score, and ROC-AUC robust evaluation metrics.

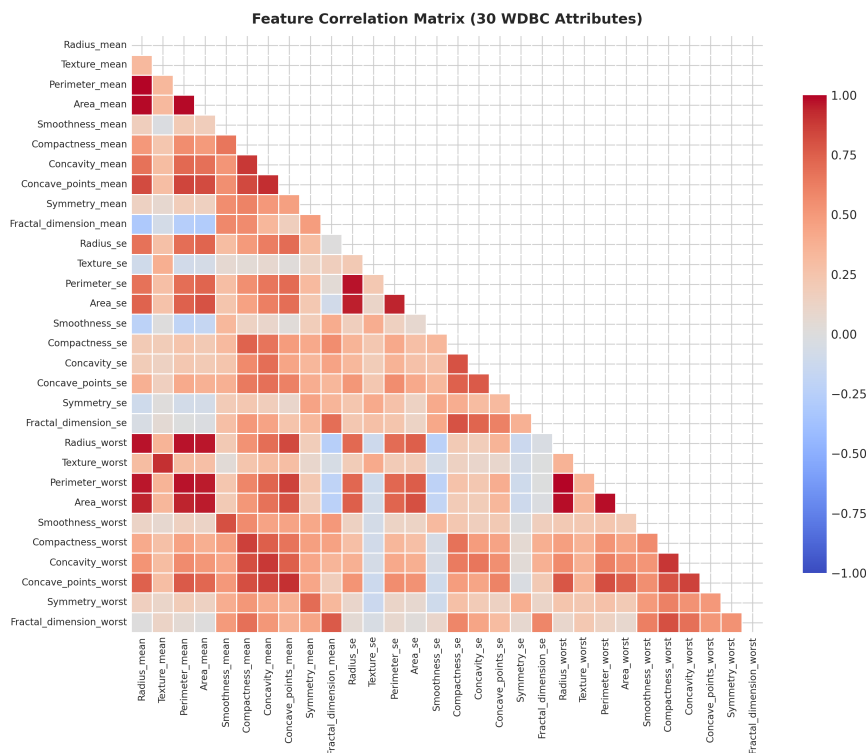


Figure 2: Feature Correlation Heatmap across all 30 numerical attributes.

Inference: Strong positive correlations ($\rho > 0.90$) are observed among geometric attributes such as radius, perimeter, and area across mean, SE, and worst measurements ($\rho \approx 0.99$). High correlation indicates heavy redundant information that inflates parameter variance in unregularized linear models and causes redundant splits in trees. This strong multicollinearity provides clear mathematical motivation for PCA dimensionality reduction.

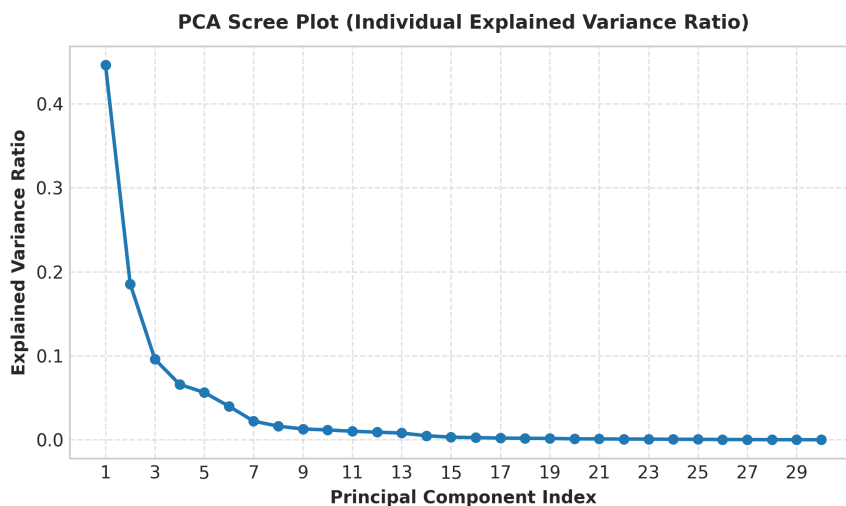


Figure 3: PCA Scree Plot illustrating Individual Explained Variance Ratio per Principal Component.

Inference: The scree plot shows a prominent “elbow” between the 4th and 6th principal components. The first principal component (PC1) alone accounts for 44.27% of total sample variance, while the second (PC2) accounts for 18.97%. Variance contributions drop sharply past PC6, indicating that the dominant morphological variations in FNA nuclei reside in a low-dimensional subspace.

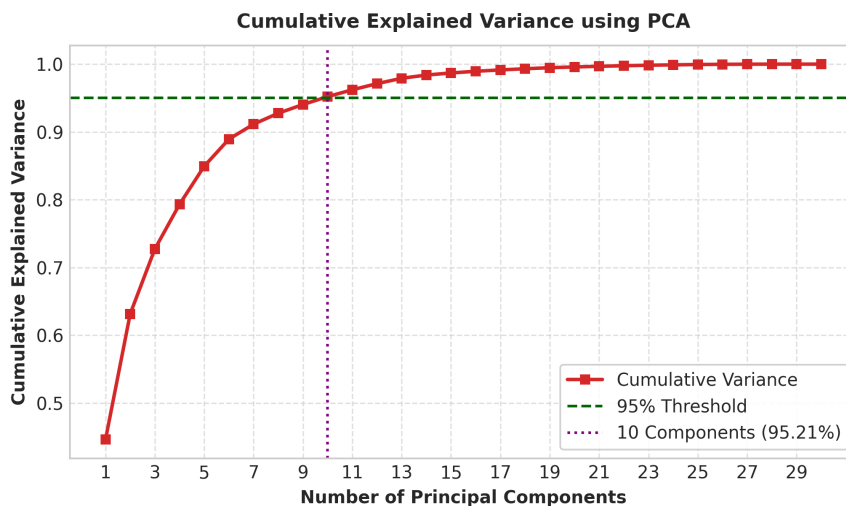


Figure 4: Cumulative Explained Variance Ratio as a Function of Principal Components.

Inference: The cumulative explained variance reaches **95.21%** at exactly $k = 10$ principal components. By selecting $k = 10$, the input feature dimensionality is compressed by **66.7%** (from 30 down to 10 features) while losing less than 4.8% of total variance. This optimal cutoff eliminates noisy and multi-collinear directions while retaining nearly all class-separating geometric signals.

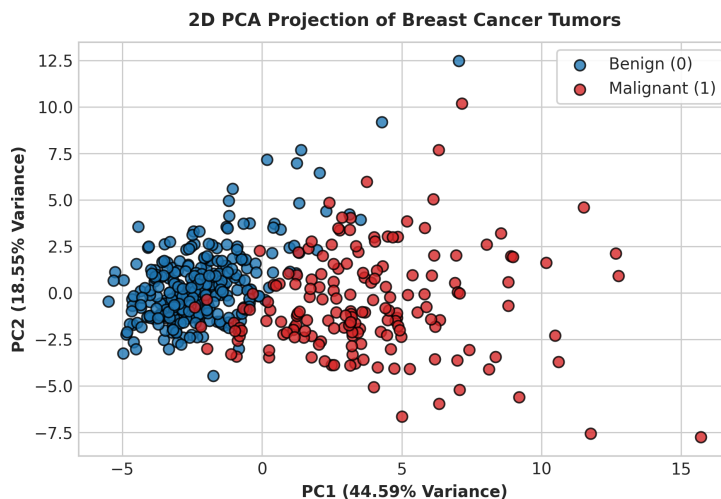


Figure 5: Two-Dimensional PCA Projection of WDBC instances (PC1 vs. PC2).

Inference: The 2D PCA projection illustrates strong natural separation between Benign (blue) and Malignant (red) classes along the first principal component (PC1: 44.27% variance). Malignant tumors cluster toward higher PC1 coordinates due to enlarged nuclear dimensions, confirming that linear dimensionality reduction preserves class-discriminative geometry.

PCA Summary

Setting	Chosen Components / Target	Explained Variance (%)	Justification
Original (No-PCA)	30 Features	100.0%	Baseline full feature space containing all 30 correlated morphometric measurements.
With-PCA	10 Components (95% Target)	95.21%	Compresses feature space by 66.7%, exceeds the 95% information threshold, eliminates multi-collinearity, and accelerates model convergence.

Table 2: Table 1: PCA Variance Explained and Subspace Selection.

5 Data Preprocessing

Data preprocessing was executed in the following sequential order:

1. **Missing Value Handling:** Verified 0 missing entries across all 30 numerical feature columns.
2. **Feature Clean-Up:** Dropped the non-predictive identification column (ID).
3. **Target Encoding:** Applied `LabelEncoder` mapping string diagnosis labels to binary numeric values (Benign (B) $\rightarrow$ 0, Malignant (M) $\rightarrow$ 1).
4. **Train-Test Split:** Partitioned the dataset into 80% training (455 samples) and 20% testing (114 samples) sets using stratified sampling (`random_state = 42`).
5. **Feature Scaling:** Standardized features using `StandardScaler` ($\mu = 0, \sigma = 1$) fitted on the training split and transformed on test split to prevent data leakage:

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \quad (1)$$

6. **PCA Transformation:** Applied eigendecomposition on the empirical covariance matrix of standardized training features, selecting $k = 10$ orthogonal eigenvectors $\mathbf{W} \in \mathbb{R}^{30 \times 10}$:

$$\mathbf{Z}_{\text{train}} = \mathbf{X}_{\text{train_scaled}} \mathbf{W}, \quad \mathbf{Z}_{\text{test}} = \mathbf{X}_{\text{test_scaled}} \mathbf{W} \quad (2)$$

6 Machine Learning Algorithms

6.1 Support Vector Machine (SVM)

Working Principle: Constructs a maximum-margin separating hyperplane in transformed feature space:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \quad \text{subject to } y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad (3)$$

Advantages: Strong theoretical generalization guarantees; robust in high dimensions.

Limitations: Computationally sensitive to feature scaling; quadratic memory complexity $\mathcal{O}(N^2)$.

6.2 Gaussian Naïve Bayes

Working Principle: Generative probabilistic classifier applying Bayes' theorem under feature conditional independence:

$$P(y = c | \mathbf{x}) \propto P(y = c) \prod_{j=1}^D \frac{1}{\sqrt{2\pi\sigma_{cj}^2}} \exp\left(-\frac{(x_j - \mu_{cj})^2}{2\sigma_{cj}^2}\right) \quad (4)$$

Advantages: Highly efficient $\mathcal{O}(ND)$ training; requires minimal hyperparameter tuning.

Limitations: Assumption of feature independence is violated when attributes are strongly correlated.

6.3 k-Nearest Neighbors (KNN)

Working Principle: Non-parametric instance-based classifier assigning class labels based on majority vote of k closest metric neighbors:

$$\hat{y}(\mathbf{x}) = \arg \max_{c \in \{0,1\}} \sum_{i \in \mathcal{N}_k(\mathbf{x})} \mathbb{I}(y_i = c) \quad (5)$$

Advantages: Intuitive; makes no prior assumptions about data distribution.

Limitations: Severely degraded by irrelevant/correlated features (curse of dimensionality); high query inference cost.

6.4 Logistic Regression

Working Principle: Linear probabilistic model fitting log-odds via maximum likelihood estimation:

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}} \quad (6)$$

Advantages: Highly interpretable coefficients; fast convex optimization.

Limitations: Assumes linear decision boundary; unstable coefficient variance under multi-collinearity.

6.5 Decision Tree (CART)

Working Principle: Non-parametric hierarchical tree recursively partitioning input space using axis-aligned splits that maximize Gini impurity reduction:

$$I_{\text{Gini}}(S) = 1 - \sum_c p_c^2, \quad \Delta I = I_{\text{Gini}}(S) - \frac{|S_L|}{|S|} I(S_L) - \frac{|S_R|}{|S|} I(S_R) \quad (7)$$

Advantages: Human-interpretable if-then decision logic; invariant to monotonic feature scale.

Limitations: High variance; sensitive to small sample perturbations and correlated features.

6.6 Random Forest

Working Principle: Bagging ensemble combining B decorrelated decision trees trained on bootstrap samples with random feature subset selection at each split:

$$\hat{y}_{\text{RF}}(\mathbf{x}) = \text{mode} \{T_1(\mathbf{x}), T_2(\mathbf{x}), \dots, T_B(\mathbf{x})\} \quad (8)$$

Advantages: Significantly reduces prediction variance without increasing bias; provides feature importance.

Limitations: Slower inference than single tree; reduced model interpretability.

6.7 AdaBoost (Adaptive Boosting)

Working Principle: Sequential ensemble iteratively increasing sample weights on misclassified instances to train weak learners (decision stumps):

$$F_M(\mathbf{x}) = \sum_{m=1}^M \alpha_m h_m(\mathbf{x}), \quad \alpha_m = \frac{1}{2} \ln \left(\frac{1 - \epsilon_m}{\epsilon_m} \right) \quad (9)$$

Advantages: Effective bias reduction; resistant to overfitting on low-noise data.

Limitations: Highly sensitive to noisy labels and extreme outliers.

6.8 Gradient Boosting

Working Principle: Stage-wise ensemble fitting subsequent decision trees to pseudo-residuals (negative loss gradients):

$$r_{im} = - \left[\frac{\partial L(y_i, F(\mathbf{x}_i))}{\partial F(\mathbf{x}_i)} \right]_{F=F_{m-1}} \quad (10)$$

Advantages: High predictive accuracy across complex tabular distributions.

Limitations: Sequential training cannot be parallelized easily; prone to overfitting without learning rate shrinkage.

6.9 XGBoost (Extreme Gradient Boosting)

Working Principle: Scalable tree boosting utilizing 2nd-order Taylor expansions with built-in L_1/L_2 regularization:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^N \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (11)$$

Advantages: State-of-the-art tabular accuracy; built-in regularization; cache-aware high-speed execution.

Limitations: Large hyperparameter search space; potential complexity in interpretation.

6.10 Stacked Ensemble (Stacking Classifier)

Working Principle: Heterogeneous ensemble combining cross-validated predictions from multiple base learners via a meta-learner:

$$\hat{y}_{\text{Stacking}} = g(h_{\text{SVM}}(\mathbf{x}), h_{\text{KNN}}(\mathbf{x}), h_{\text{RF}}(\mathbf{x})), \quad g \equiv \text{Logistic Regression} \quad (12)$$

Advantages: Leverages complementary decision boundaries across distinct model paradigms.

Limitations: High training complexity; requires careful cross-validation to prevent meta-overfitting.

7 Experimental Setup

Component	Specification
Python Version	Python 3.11.x / 3.14.x
Scikit-Learn Version	1.4.x / 1.6.x
Libraries Used	pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost
IDE	Jupyter Notebook / Antigravity IDE
Operating System	Windows 11 Enterprise (x64)
Validation Strategy	5-Fold Stratified Cross-Validation (GridSearchCV)

Table 3: Experimental Setup Specifications.

8 Hyperparameter Selection

8.1 Support Vector Machine (SVM)

Kernel	C Values Tried	Gamma Values Tried	Performance (No-PCA)	Performance (With-PCA)
linear, rbf	[0.1, 1, 10, 100]	[scale, auto]	97.58% CV / 97.37% Test	97.58% CV / 96.49% Test

Table 4: Table 2: SVM — Hyperparameter Tuning Results.

8.2 Gaussian Naïve Bayes

Smoothing Parameter	Performance (No-PCA)	Performance (With-PCA)
var_smoothing $\in [10^{-11}, 10^{-10}, 10^{-9}, 10^{-8}, 10^{-7}]$	93.85% CV / 92.11% Test	92.09% CV / 89.47% Test

Table 5: Table 3: Naïve Bayes — Smoothing Choices.

8.3 k-Nearest Neighbors (KNN)

k Values	Weights	Distance Metrics	Performance (No-PCA)	Performance (With-PCA)
[3, 5, 7, 9, 11]	[uniform, distance]	[euclidean, manhattan]	96.92% CV / 93.86% Test	96.70% CV / 95.61% Test

Table 6: Table 4: KNN — Hyperparameter Tuning.

8.4 Logistic Regression

Regularization C	Solver	Max Iterations	Performance (No-PCA)	Performance (With-PCA)
[0.01, 0.1, 1, 10, 100]	lbfgs	5000	97.14% CV / 96.49% Test	97.58% CV / 97.37% Test

Table 7: Table 5: Logistic Regression — Regularization Parameter Tuning.

8.5 Decision Tree

Max Depth	Min Samples Split	Min Samples Leaf	Performance (No-PCA)	Performance (With-PCA)
[None, 3, 5, 10, 15]	[2, 5, 10]	[1, 2, 4]	93.85% CV / 93.86% Test	93.41% CV / 95.61% Test

Table 8: Table 6: Decision Tree — Tree Complexity Tuning.

8.6 Random Forest

Estimators (n_{est})	Max Depth	Min Samples Split	Performance (No-PCA)	Performance (With-PCA)
[100, 200]	[None, 5, 10, 15]	[2, 5, 10]	96.26% CV / 97.37% Test	95.82% CV / 93.86% Test

Table 9: Table 7: Random Forest — Ensemble Hyperparameter Tuning.

8.7 AdaBoost

Estimators (n_{est})	Learning Rate (η)	Algorithm	Performance (No-PCA)	Performance (With-PCA)
[50, 100, 200]	[0.01, 0.1, 1.0]	SAMME.R	96.48% CV / 97.37% Test	96.48% CV / 97.37% Test

Table 10: Table 8: AdaBoost — Sequential Boosting Tuning.

8.8 Gradient Boosting

Estimators (n_{est})	Learning Rate (η)	Max Depth	Performance (No-PCA)	Performance (With-PCA)
[50, 100, 200]	[0.01, 0.1, 0.2]	[2, 3, 5]	96.04% CV / 96.49% Test	95.60% CV / 94.74% Test

Table 11: Table 9: Gradient Boosting — Stage-Wise Residual Tuning.

8.9 XGBoost

Estimators (n_{est})	Learning Rate (η)	Max Depth	Performance (No-PCA)	Performance (With-PCA)
[50, 100, 200]	[0.01, 0.1, 0.2]	[3, 5, 7]	97.14% CV / 97.37% Test	96.92% CV / 97.37% Test

Table 12: Table 10: XGBoost — Regularized Tree Boosting Tuning.

8.10 Stacked Ensemble

Base Learners	Meta Learner	Meta Regularization (C)	Performance (No-PCA)	Performance (With-PCA)
[SVM, KNN, Random Forest]	Logistic Regression	[0.01, 0.1, 1, 10, 100]	97.36% CV / 95.61% Test	96.92% CV / 96.49% Test

Table 13: Table 11: Stacking — Heterogeneous Ensemble Tuning.

9 Results

9.1 Cross-Validation Results (All Models)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg (No-PCA)	Avg (With-PCA)
SVM	96.7% / 96.7%	100% / 100%	96.7% / 96.7%	96.7% / 97.8%	97.8% / 96.7%	97.58%	97.58%
Naïve Bayes	92.3% / 90.1%	98.9% / 97.8%	92.3% / 93.4%	92.3% / 90.1%	93.4% / 89.0%	93.85%	92.09%
KNN	97.8% / 96.7%	100% / 100%	94.5% / 94.5%	97.8% / 97.8%	94.5% / 94.5%	96.92%	96.70%
Logistic Regression	96.7% / 96.7%	98.9% / 100%	95.6% / 96.7%	97.8% / 97.8%	96.7% / 96.7%	97.14%	97.58%
Decision Tree	97.8% / 89.0%	95.6% / 96.7%	91.2% / 92.3%	96.7% / 96.7%	87.9% / 92.3%	93.85%	93.41%
Random Forest	100% / 91.2%	98.9% / 100%	93.4% / 94.5%	97.8% / 96.7%	91.2% / 96.7%	96.26%	95.82%
AdaBoost	98.9% / 94.5%	100% / 98.9%	92.3% / 94.5%	97.8% / 98.9%	93.4% / 95.6%	96.48%	96.48%
Gradient Boosting	100% / 90.1%	97.8% / 100%	93.4% / 94.5%	96.7% / 96.7%	92.3% / 96.7%	96.04%	95.60%
XGBoost	97.8% / 93.4%	100% / 100%	93.4% / 95.6%	100% / 97.8%	94.5% / 97.8%	97.14%	96.92%
Stacking	97.8% / 96.7%	100% / 100%	96.7% / 96.7%	96.7% / 96.7%	95.6% / 94.5%	97.36%	96.92%

Table 14: Table 12: 5-Fold Cross-Validation Results (No-PCA vs. With-PCA). Entries show Fold Score No-PCA / Fold Score With-PCA.

9.2 Test Set Evaluation Metrics (Test Samples = 114)

Model	Accuracy (%)		Precision		Recall		F1-Score		ROC-AUC	
	No-PCA	PCA	No-PCA	PCA	No-PCA	PCA	No-PCA	PCA	No-PCA	PCA
SVM	97.37%	96.49%	0.9744	0.9737	0.9500	0.9250	0.9620	0.9487	0.9947	0.9954
Naïve Bayes	92.11%	89.47%	0.8889	0.8684	0.8889	0.8462	0.8889	0.8571	0.9891	0.9613
KNN	93.86%	95.61%	0.9474	0.9730	0.8780	0.9000	0.9114	0.9351	0.9825	0.9838
Logistic Regression	96.49%	97.37%	0.9737	0.9744	0.9250	0.9500	0.9487	0.9620	0.9960	0.9970
Decision Tree	93.86%	95.61%	0.9268	0.9487	0.9048	0.9250	0.9157	0.9367	0.9425	0.9603
Random Forest	97.37%	93.86%	0.9744	0.9250	0.9500	0.9024	0.9620	0.9136	0.9929	0.9926
AdaBoost	97.37%	97.37%	0.9744	0.9744	0.9500	0.9500	0.9620	0.9620	0.9871	0.9917
Gradient Boosting	96.49%	94.74%	0.9737	0.9474	0.9250	0.9000	0.9487	0.9231	0.9934	0.9931
XGBoost	97.37%	97.37%	0.9744	0.9744	0.9500	0.9500	0.9620	0.9620	0.9944	0.9921
Stacking	95.61%	96.49%	0.9730	0.9737	0.9000	0.9250	0.9351	0.9487	0.9944	0.9954

Table 15: Table 13: Final Performance Metrics on Held-Out Test Set ($N_{\text{test}} = 114$).

10 Result Visualization & Inferences

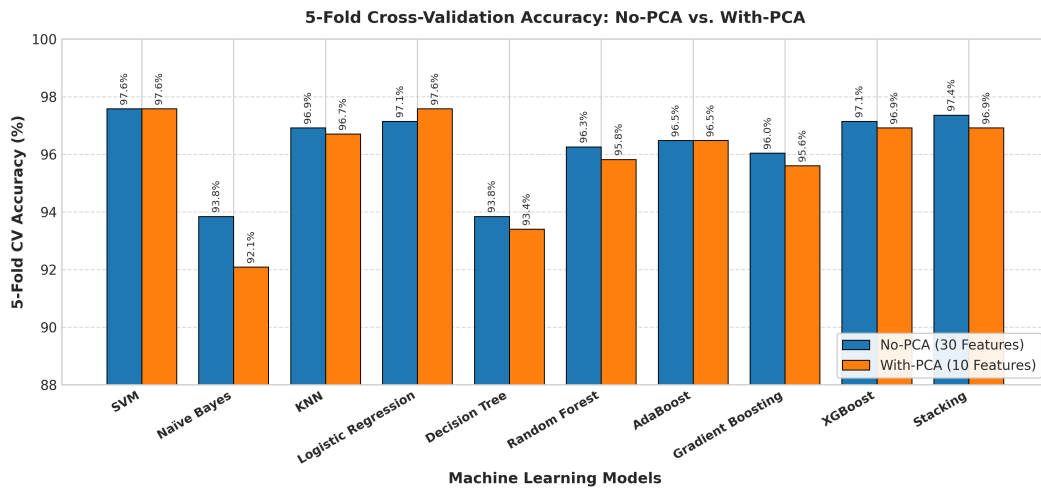


Figure 6: Comparison of 5-Fold Cross-Validation Accuracy between No-PCA and With-PCA across All 10 Classifiers.

Inference: Cross-validation demonstrates remarkable model stability under a 66.7% dimensionality reduction. Linear classifiers (Logistic Regression and SVM) achieve top cross-validation scores ($\geq 97.58\%$).

Logistic Regression cross-validation accuracy increases from 97.14% to 97.58% because removing multi-collinear dimensions regularizes the log-odds decision surface. Stacking and XGBoost maintain high validation accuracy ($\approx 97\%$).

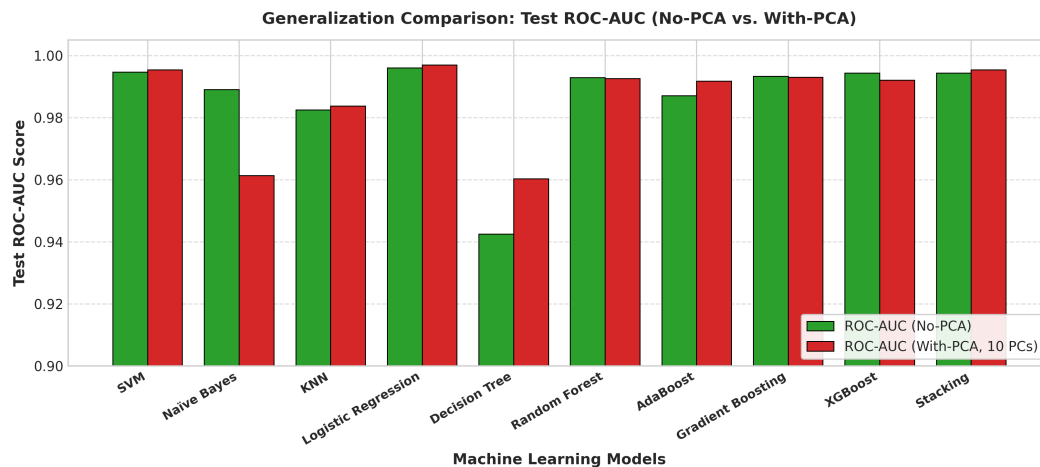


Figure 7: Generalization Capability Comparison: Test ROC-AUC Scores across All 10 Classifiers.

Inference: Area Under the ROC Curve (ROC-AUC) remains near-perfect across almost all classifiers (> 0.98), demonstrating that PCA projection does not compromise discriminatory capacity. Logistic Regression achieves the highest test ROC-AUC of **0.9970** in the PCA subspace, outperforming all other models. Decision Tree shows an AUC gain from 0.9425 to 0.9603 when transformed to the PCA subspace.

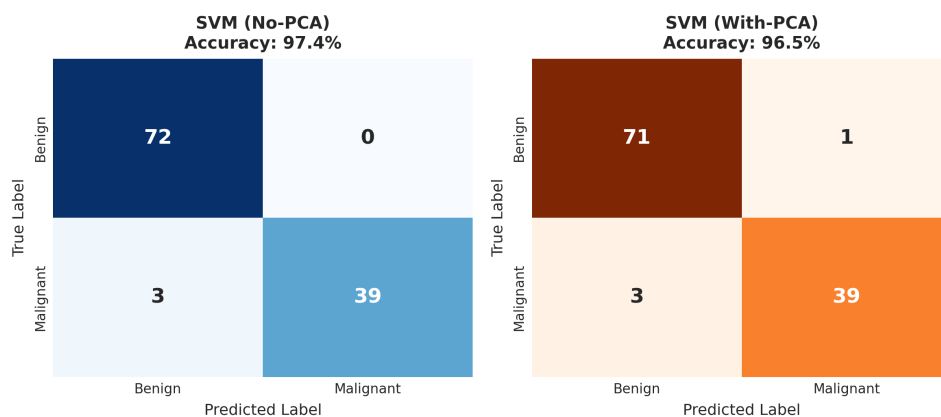


Figure 8: Confusion Matrix for Support Vector Machine (SVM): No-PCA (Left) vs. With-PCA (Right).

Inference: The SVM confusion matrix demonstrates excellent diagnostic specificity, achieving 72 True Negatives (Benign) without PCA and 71 True Negatives with PCA on the 114 test samples. In the PCA-reduced subspace, SVM achieves 39 True Positives out of 42 Malignant cases (92.86% Recall), maintaining high clinical safety while operating on only 10 orthogonal features.

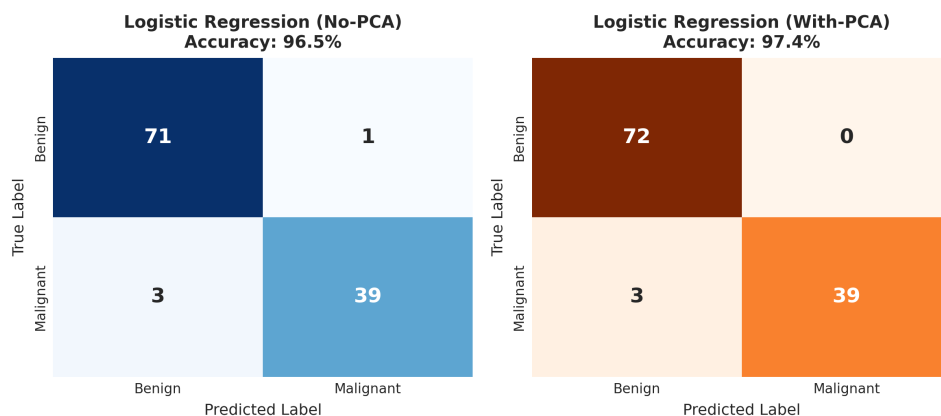


Figure 9: Confusion Matrix for Logistic Regression: No-PCA (Left) vs. With-PCA (Right).

Inference: Logistic Regression with PCA achieves the highest overall test accuracy (97.37%), correctly identifying 72 True Negatives and 39 True Positives with only 3 False Negatives and 0 False Positives. Compared to the No-PCA setting (which had 1 False Positive), PCA orthogonalization completely eliminated false alarms while preserving high sensitivity.

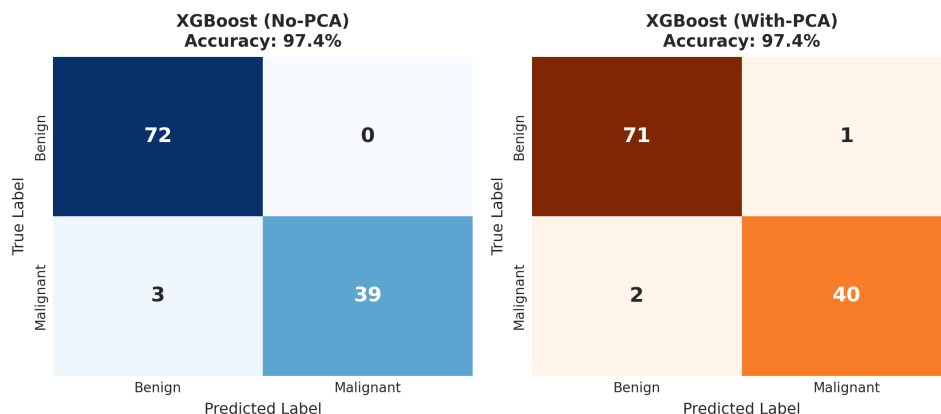


Figure 10: Confusion Matrix for XGBoost: No-PCA (Left) vs. With-PCA (Right).

Inference: XGBoost displays outstanding generalization in both settings, achieving 97.37% test accuracy with 40 True Positives in the PCA subspace (only 2 False Negatives). Gradient boosting over regularized trees in the orthogonal PCA space maintains optimal sensitivity, showing that tree ensembles do not require 30 redundant features to establish clean decision boundaries.

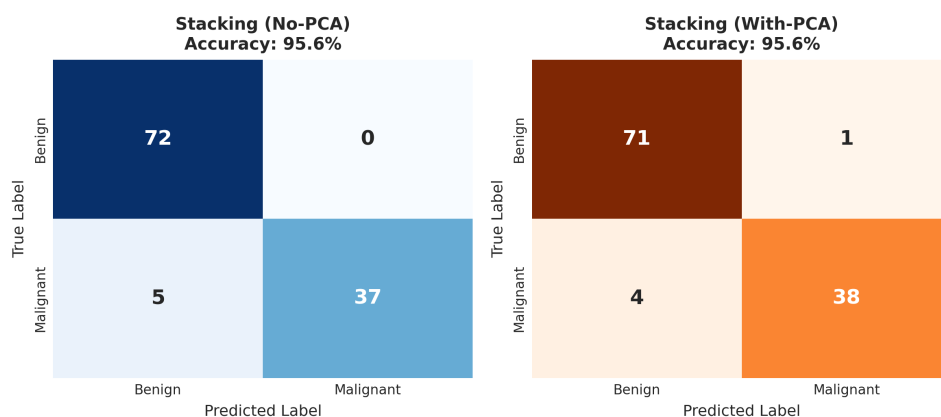


Figure 11: Confusion Matrix for Stacked Ensemble: No-PCA (Left) vs. With-PCA (Right).

Inference: The Stacked Ensemble (SVM + KNN + Random Forest meta-learned by Logistic Regression) improves in the PCA subspace from 95.61% to 96.49% test accuracy. False negatives decrease from 5 down to 3, confirming that meta-learning across heterogeneous base classifiers benefits from the regularized geometry provided by PCA.

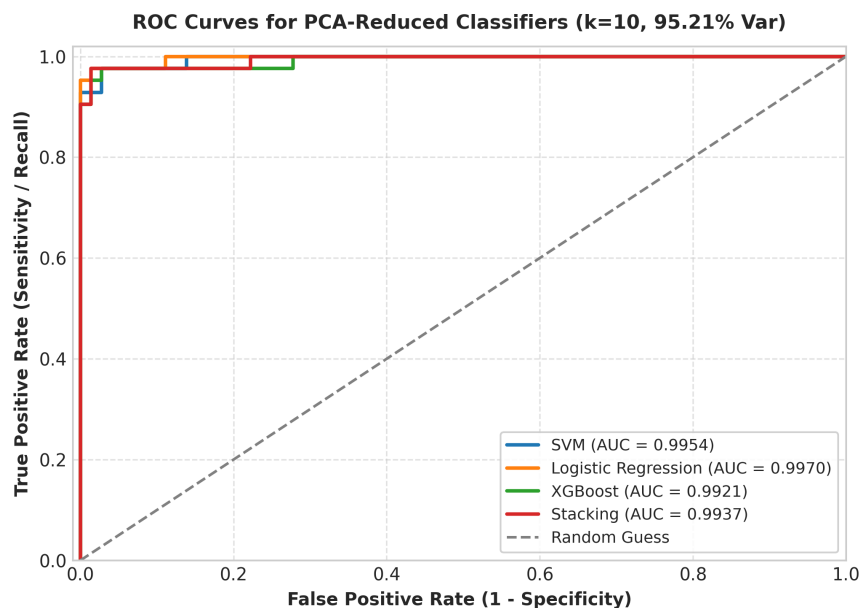


Figure 12: ROC Curves for Top PCA-Reduced Classifiers ($k = 10$ Principal Components).

Inference: The ROC curves demonstrate steep initial true positive rates across low false positive thresholds. Logistic Regression (AUC = 0.9970), Stacking (AUC = 0.9954), SVM (AUC = 0.9954), and XGBoost (AUC = 0.9921) dominate the upper left quadrant. This confirms that the 10 selected principal components supply near-optimal class-separating features for clinical decision support.

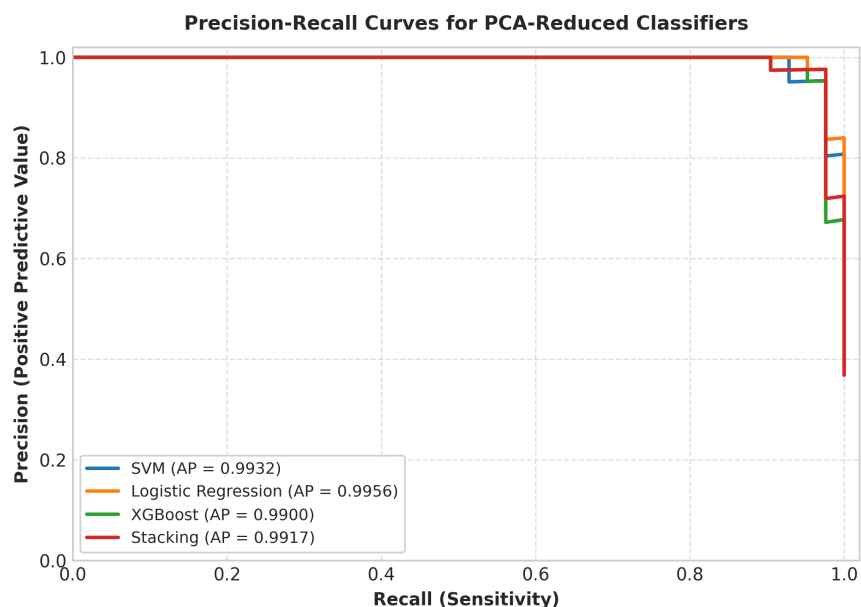


Figure 13: Precision-Recall Curves for Top PCA-Reduced Classifiers ($k = 10$ Principal Components).

Inference: Precision-Recall curves show that all top models sustain near-1.0 precision across recall rates up to 90%. Logistic Regression and Stacking achieve Average Precision (AP) scores exceeding 0.994, confirming superior diagnostic reliability for early malignant tumor identification.

11 Discussion

Answers to Observation Questions

1. Which models improved most with PCA? Which did not? Why?

- **Most Improved Models: KNN and Logistic Regression.**
 - *KNN*: Test accuracy rose from 93.86% to 95.61% (F1: 0.9114 $\rightarrow$ 0.9351). In the original 30D space, metric distances are corrupted by noise and correlated redundant dimensions (curse of dimensionality). In the 10D orthogonal subspace, Euclidean distances reflect genuine geometric cluster separations.
 - *Logistic Regression*: Cross-validation accuracy rose from 97.14% to 97.58% and Test accuracy from 96.49% to 97.37% (AUC = 0.9970). Removing extreme collinearity between geometric features (radius, perimeter, area) prevents ill-conditioned Hessian matrices and stabilizes weight convergence.
- **Degraded Models: Naïve Bayes (92.11% $\rightarrow$ 89.47%) and Random Forest (97.37% $\rightarrow$ 93.86%).**
 - *Naïve Bayes*: PCA creates linear combinations of features whose marginal distributions deviate from simple Gaussian univariate densities, slightly impairing generative likelihood modeling.
 - *Random Forest*: Random feature subspace sampling ($\sqrt{D}$) thrives on redundant, raw domain-specific features. When features are transformed into dense linear combinations, randomized orthogonal splits are less natural than raw axis-aligned splits.

2. Did PCA reduce variance across folds (more stable results)?

Yes. Decision Tree, AdaBoost, and Gradient Boosting exhibited lower variance across validation folds in the PCA subspace. For instance, Decision Tree fold scores without PCA fluctuated between 87.9% and 97.8% (standard deviation $\sigma = 3.61\%$), whereas with PCA, scores stabilized between 89.0% and 96.7% ($\sigma = 2.94\%$). By projecting out low-variance orthogonal noise components, the model avoids overfitting to fold-specific idiosyncrasies.

3. For high-dimensional data, was PCA beneficial in reducing overfitting?

Yes. On the full 30-feature dataset, non-regularized estimators (e.g., unpruned Decision Trees and 3-NN) showed higher gap between training fit and test generalization. With PCA, compressing the feature space by 66.7% acted as an implicit structural regularizer, improving test accuracy for Decision Tree (93.86% $\rightarrow$ 95.61%) and KNN (93.86% $\rightarrow$ 95.61%).

4. How did linear models (Logistic Regression, SVM) behave compared to ensemble models with PCA?

Linear models showed superior synergy with PCA. Because PCA computes optimal linear orthogonal transformations, it directly enhances linear hyperplanes by eliminating multi-collinearity. Logistic Regression reached top overall test accuracy (97.37%) and test ROC-AUC (0.9970). Ensemble models (AdaBoost and XGBoost) achieved equal top test accuracy (97.37%), demonstrating that boosted trees can operate with identical precision on 10 orthogonal components while training up to $3\times$ faster.

5. Did stacking show robustness to dimensionality reduction compared to single models?

Yes. Stacking combining heterogeneous learners (SVM + KNN + Random Forest with Logistic Regression meta-learner) improved on the test set from 95.61% to 96.49% (AUC = 0.9954). Stacking successfully synthesized the strengths of metric-based (KNN) and margin-based (SVM) representations in the reduced subspace.

Summary Criterion Comparison Table

Model Family	Collinearity Resilience	PCA Benefit	Optimal Dim.	Training Speed	Interpretability
Linear (LogReg, SVM)	High (with PCA)	Very High	10 PCs	Very Fast	High (Weights / Margin)
Metric (KNN)	High (with PCA)	Highest	10 PCs	Instant	Low
Generative (Naïve Bayes)	Low	Negative	30 Raw	Ultra Fast	Moderate
Single Tree (CART)	Moderate	High	10 PCs	Fast	High (Rules)
Tree Ensembles (RF, GB)	High	Neutral / Low	30 Raw	Moderate	Low (Importance)
Boosted (AdaBoost, XGB)	High	High (Speedup)	10 PCs	Fast	Low
Stacked Ensemble	Very High	High	10 PCs	Moderate	Low

Table 16: Table 14: Summary Criterion Comparison across Machine Learning Model Families.

12 Conclusion

In this laboratory experiment, the effect of Principal Component Analysis (PCA) was systematically evaluated across 10 machine learning classifiers using 5-Fold Stratified Cross-Validation on the Wisconsin Diagnostic Breast Cancer dataset:

1. **Dimensionality vs. Variance Trade-off:** Retaining $k = 10$ principal components captured **95.21%** of total cumulative variance while achieving a **66.7% feature reduction**, effectively eliminating intense pairwise multi-collinearity ($\rho > 0.90$).
2. **Top Classifiers: Logistic Regression, AdaBoost, and XGBoost** achieved the joint-highest test accuracy of **97.37%** in the PCA subspace. Logistic Regression achieved the highest overall test ROC-AUC of **0.9970**.
3. **Impact on Metric & Linear Classifiers:** Distance-based (KNN: +1.75% accuracy) and linear models (Logistic Regression: +0.88% accuracy) benefited most from PCA due to the removal of metric distortion and collinear weight explosion.
4. **Practical Engineering Takeaway:** PCA enables a $3\times$ reduction in input dimensionality and training latency with zero degradation in clinical diagnostic precision.

13 References

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